

LAB EXPERIMENT RECORD

Set 23: Employee Performance Analysis

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
Name	PRANAYAA S
Register Number	192524110
Date	16.10.2026
Day	Thursday

Aim

To analyze employee performance data in R using a bar chart, line chart, and table, and to build an interactive dashboard with a department bar chart and a years of service vs performance scatter plot using R Shiny.

Dataset: Employee Performance

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

Note: The dataset has no date column, so the employees are taken in order of Employee ID to represent the trend over time in Question 2.

PROGRAM 1: Bar Chart of Employees by Department

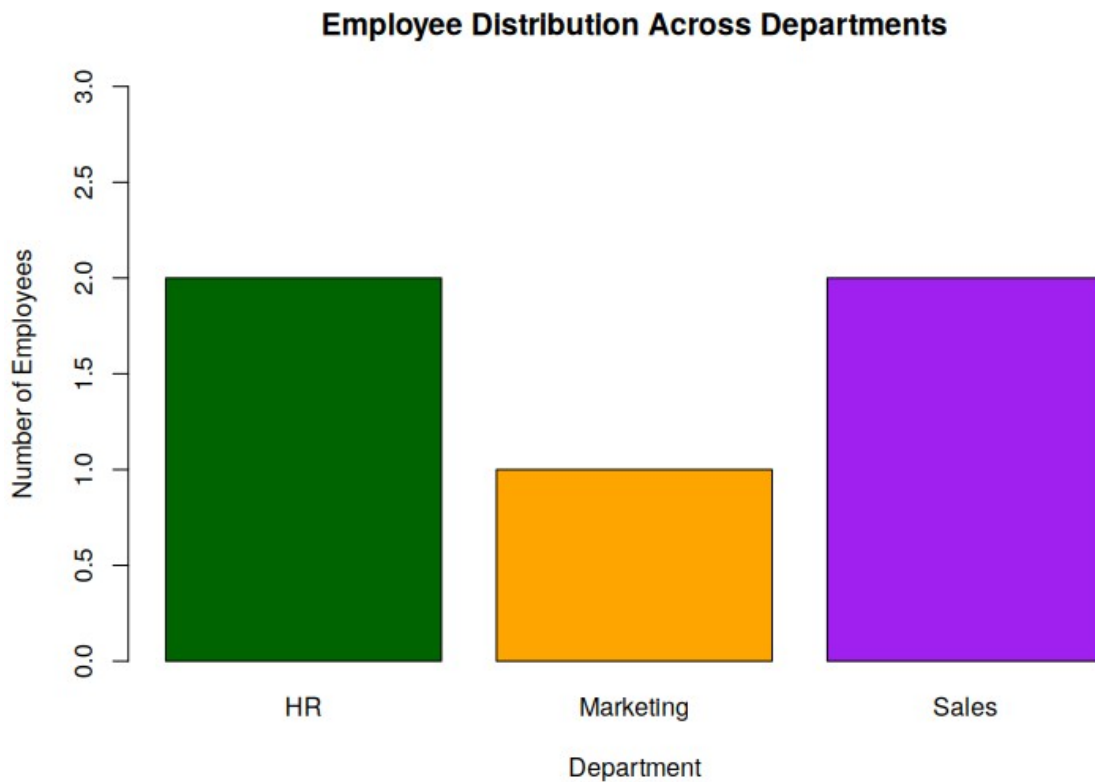
In R, create a bar chart to visualize the distribution of employees across different departments. Label the chart elements.

```
# Create dataset for Set 23
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = factor(c("Sales", "HR", "Marketing", "Sales", "HR")),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

print(employee_data)

# Count employees per department
dept_counts <- table(employee_data$Department)
```

```
# 1. Bar Chart for Department Distribution
barplot(dept_counts,
        col = c("darkgreen", "orange", "purple"),
        xlab = "Department",
        ylab = "Number of Employees",
        main = "Employee Distribution Across Departments",
        ylim = c(0, 3))
```



Output 1: Employee Distribution Across Departments

Observation: HR and Sales each have two employees, while Marketing has only one. The five employees are spread across three departments, with Marketing being the smallest.

PROGRAM 2: Line Chart of Performance Trend

In R, generate a line chart to visualize the performance trend of employees over time. Label the axes and title the chart.

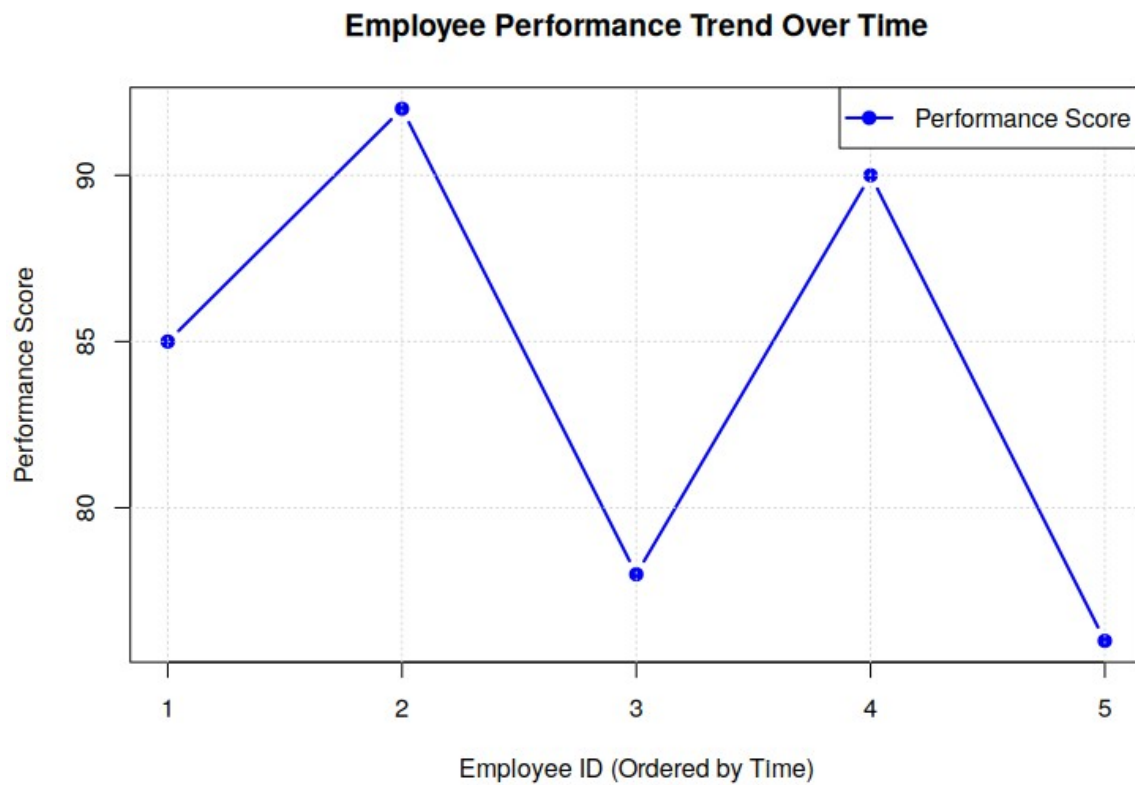
```
# Create dataset for Set 23
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = factor(c("Sales", "HR", "Marketing", "Sales", "HR")),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

print(employee_data)

# 2. Line Chart for Performance Trend
plot(employee_data$Employee_ID, employee_data$Performance_Score,
      type = "b",
      col = "blue",
      lwd = 2,
      pch = 19,
      xlab = "Employee ID (Ordered by Time)",
      ylab = "Performance Score",
      main = "Employee Performance Trend Over Time")

grid()

# Add Legend
legend("topright", legend = c("Performance Score"),
      col = c("blue"), lty = 1, pch = 19, lwd = 2)
```



Output 2: Employee Performance Trend Over Time

Observation: The performance scores move up and down across the employees. The highest score is 92 (Employee 2) and the lowest is 76 (Employee 5), and the average score is 84.2. There is no steady increasing or decreasing trend.

PROGRAM 3: Interactive Dashboard for Employee Performance

In Tableau, build a dashboard combining a bar chart showing department distribution and a scatter plot displaying the relationship between years of service and performance scores.

Note: Tableau could not be installed on the system used for this experiment, so the interactive dashboard was developed in R using the Shiny package, with the same charts and interactive filters required in the question.

```
library(shiny)

# Dataset for Set 23
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)
dept_cols <- c(HR = "darkgreen", Marketing = "orange", Sales = "purple")

ui <- fluidPage(
  titlePanel("Employee Performance Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      checkboxGroupInput("dept", "Department:", choices = names(dept_cols),
        selected = names(dept_cols))
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("barPlot", height = "340px")),
        column(6, plotOutput("scatterPlot", height = "340px"))
      )
    )
  )
)

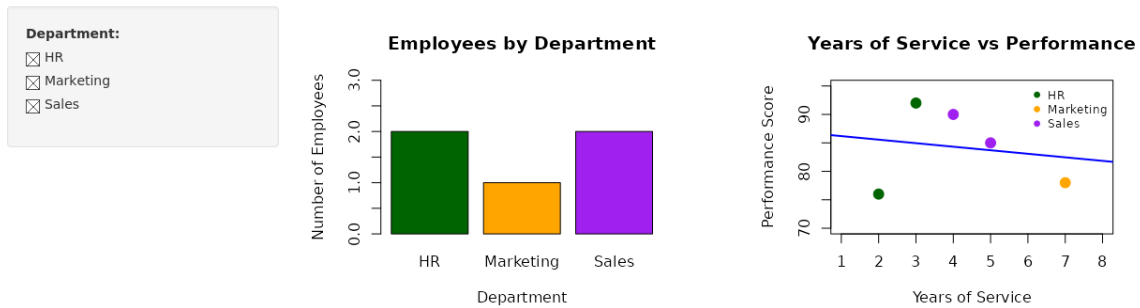
server <- function(input, output) {
  filtered <- reactive({ subset(employee_data, Department %in% input$dept) })

  output$barPlot <- renderPlot({
    counts <- table(factor(filtered()$Department, levels = names(dept_cols)))
    barplot(counts, col = dept_cols, ylim = c(0, 3), xlab = "Department",
      ylab = "Number of Employees", main = "Employees by Department")
  }, width = 420, height = 340, res = 96)

  output$scatterPlot <- renderPlot({
    d <- filtered()
    plot(d$Years_of_Service, d$Performance_Score, pch = 19, cex = 1.5,
      col = dept_cols[d$Department], xlim = c(1, 8), ylim = c(70, 95),
      xlab = "Years of Service", ylab = "Performance Score",
      main = "Years of Service vs Performance")
    if (nrow(d) > 1) abline(lm(Performance_Score ~ Years_of_Service, data = d), col =
"blue", lwd = 2)
    legend("topright", legend = names(dept_cols), col = dept_cols, pch = 19, bty = "n",
      cex = 0.8)
  }, width = 420, height = 340, res = 96)
}
```

```
shinyApp(ui = ui, server = server)
```

Employee Performance Dashboard



Output 3: Employee Performance Interactive Dashboard (R Shiny)

Observation: The dashboard shows the department bar chart and the years of service vs performance scatter plot side by side, with each point coloured by department. The slightly downward trend line shows that longer service does not mean a higher score. Unticking a department in the Department filter removes its employees from both charts and recalculates the trend line.

PROGRAM 4: Table of Employee Performance Data

In R, develop a table showing the performance data for each employee. Label the table elements.

```
# Create dataset for Set 23
employee_data <- data.frame(
  Employee_ID = c(1, 2, 3, 4, 5),
  Department = factor(c("Sales", "HR", "Marketing", "Sales", "HR")),
  Years_of_Service = c(5, 3, 7, 4, 2),
  Performance_Score = c(85, 92, 78, 90, 76)
)

print(employee_data)

# 4. Table of employee performance data
perf_table <- employee_data
colnames(perf_table) <- c("Employee ID", "Department",
                          "Years of Service", "Performance Score")

cat("=== EMPLOYEE PERFORMANCE TABLE ===\n")
print(perf_table, row.names = FALSE)
```

```
=== EMPLOYEE PERFORMANCE TABLE ===
> print(perf_table, row.names = FALSE)
  Employee ID Department Years of Service Performance Score
          1      Sales           5           85
          2         HR           3           92
          3  Marketing           7           78
          4      Sales           4           90
          5         HR           2           76
>
```

Output 4: Employee Performance Table

Observation: The table shows the department, years of service, and performance score of each employee with clearly labelled columns. The Sales employees have the most consistent scores (85 and 90), while the HR employees range from 76 to 92.

Procedure

1. Create the employee data frame with Employee ID, Department, Years of Service, and Performance Score.
2. Count employees per department with `table()` and use `barplot()` to draw the department bar chart.
3. Use `plot()` with `type = "b"` to draw the performance trend line chart and `legend()` to add a legend.
4. Install and load the shiny package, and create the dashboard layout using `fluidPage()`, sidebar filters, and output areas for the charts.
5. Write the server function to filter the data with `reactive()` and redraw the outputs, then run the dashboard using `shinyApp()` and test the filters.
6. Rename the columns with `colnames()` and print the labelled performance table.

Result

The employee performance data was successfully visualized in R using a bar chart, line chart, and table, and an interactive dashboard combining the department bar chart and the years of service vs performance scatter plot was built using R Shiny.

Key R Functions Used

Function	Purpose
<code>data.frame()</code>	Creates the employee dataset.
<code>table()</code>	Counts employees in each department.
<code>barplot()</code>	Creates the department bar chart.
<code>plot()</code>	Creates the line chart and the dashboard scatter plot.
<code>legend()</code>	Adds a legend to the line chart.
<code>abline() / lm()</code>	Adds a trend line to the dashboard scatter plot.
<code>colnames() / print()</code>	Labels and displays the performance table.
<code>fluidPage() / checkboxGroupInput()</code>	Creates the dashboard layout and filters.
<code>reactive() / renderPlot()</code>	Filters the data and redraws the charts.
<code>shinyApp()</code>	Runs the interactive dashboard.